ETHNOHACK 2026 · TRACKS 01 & 02

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Everyone built a system that *reads* ancient medicine.
We built one that *argues with it*.

A 1.5 B open model · 940 MiB · runs on a CPU · everything open

# The record is unusable for two reasons — neither is data volume

**1 · The names don't survive**

**甘草 · 감초 · Glycyrrhiza uralensis**

…and also Liquorice, Mulethi. A 16th-century drug name and a modern gene target never meet.

**2 · Nobody checked the chemistry**

A preparation instruction is a chemical protocol written in a language no one validated against thermodynamics.

The binding constraint was never data. It was **translation**, and then **verification**. We built both.

# Track 01 · Paste a passage, get structured knowledge



甘草（炙）二兩、人參一兩、白朮二兩、茯苓二兩，右四味，以水六升，煮取三升，溫服一升，日三服。治脾胃虛弱，氣短之方。

Extract try: formula (Taiwan, trad.) | formula (simplified) | materia medica note

**4 drugs recognized**

| AS WRITTEN | MODERN BINOMIAL | COMPOUND CLASSES | GENE TARGETS |
|---|---|---|---|
| 甘草<br>span 0-2 · conf 0.95 | Glycyrrhiza uralensis | flavonoid, phenylpropanoid, steroid, terpenoid | ABCB1, ABCB11, ABCC1, ABCC2 |
| 人參<br>span 8-10 · conf 0.95 | Panax ginseng | terpenoid | ABAT, ABCB1, ABCB11, ABCC1 |
| 白朮<br>span 13-15 · conf 0.95 | Atractylodes macrocephala | alkaloid, flavonoid, terpenoid | ACHE, ADORA1, ADORA2A, ADORA3 |
| 茯苓<br>span 18-20 · conf 0.95 | Wolfiporia cocos | terpenoid | — |

Every drug carries a **character span and a confidence**. Extraction is rule- and lexicon-based, so it cannot invent a span. 脾胃虛弱 → **dyspepsia**.

# At corpus scale, not just on a paragraph

**273,334**
formulas

**1,082**
classical works

**744,997**
drug mentions

**714,079**
drug–indication pairs

**77,441**
indications

**52 s**
on a CPU

From 1,304 CC0 classical works. Span-level provenance on every pair. 61.3 % of formulas contain a recognised drug.

# Track 02 · Type a target, get a ranked shortlist



## Screen a target — Track 02, live

Type a disease target. **Layer 1** finds the plant→compound→target links from the knowledge graph; **layer 2** ranks our 23,176 plant compounds by ECFP4 similarity to the target's known ChEMBL actives; **layer 3** flags Lipinski ADMET, preparation physics and structural novelty. Every row is a falsifiable candidate, not a claim.

EGFR

Screen try: COX2 tubulin EGFR PTGS2

### Epidermal growth factor receptor `CHEMBL3608`

58 known actives · layer 1: plant→compound→target KG · layer 2: ECFP4 Tanimoto vs ChEMBL actives · layer 3: Lipinski ADMET + preparation physics + structural novelty

| # | COMPOUND | PLANT | SIMILARITY | ADMET | PHYSICS | FUSED |
|---|---|---|---|---|---|---|
| 1 | 2,4-Dihydroxybenzaldehyde | Cassia fistula | 0.6667 | pass | n/a | **0.667** |
| 2 | 2,4-Dihydroxybenzaldehyde | Morus alba | 0.6667 | pass | n/a | **0.667** |

23,176 plant compounds · ECFP4 vs known ChEMBL actives · Lipinski ADMET · preparation physics · novelty. **HMGCR returns Tanimoto 1.000** — the pipeline rediscovers a known active. That's our control.

# The part nobody else has

**78**/203

recorded preparation methods are **physically impossible** — the compound decomposes at the temperature the text prescribes.

And **6 of them** passed every text-based check. The ancient record agreed. The tradition agreed. Physics said no.

A system that only reads texts cannot know that. Ancient consensus is not evidence.

# The model underneath

**Qwen2.5-1.5B fine-tune**

**940 MiB** Q4_K_M · runs on a CPU

**49.0 % vs 37.0 %** over its base, 200 held-out questions

LoRA on 88,082 domain examples

**…and what lost**

4-expert sparse MoE, 1.44 B, 999 MiB

**31.7 % vs 43.3 %** — worse than dense

Released anyway, with the benchmark

# What we can't do — stated before you ask

**48.7 %**

of plant materia medica resolves. The rest are archaic names absent from every open lexicon.

**19 records**

is all the Greek and Latin text we hold. We don't claim to decode them.

**~0 %**

of the corpus is animal, mineral, or manufactured objects — 37.9 %. Counting them as failures was the wrong denominator.

We tested character-similarity fallback as a coverage fix: it recovered 2.9 % and was wrong half the time. Not shipped.

## Why it matters

A laptop in a clinic in Yunnan or Kerala can run this offline. No GPU, no cluster, no API. The corpus is enumerated, every claim cites a span, and the licence audit includes the sources we **rejected**.

The contribution isn't a bigger model. It's a system that links eras — and then checks itself.

# Try it

**Live demo** · tomscot-shennong.hf.space

**Code** · github.com/cptlight/shennong

**Weights** · huggingface.co/tomscot/shennong-1.5b-sft

14 tabs, 4 running live inference on CPU. 78 of 203 claims refuted. Everything open, permissive licences only.